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Ex.No:4 **Date:1/9/25**

PACKET SNIFFING USING RAW SOCKETS IN PYTHON

Aim:

To implement packet sniffing using raw sockets in python that captures Ethernet frames and displays source/destination MACs and protocol info for network analysis and troubleshooting.

Introduction:

Packet Sniffing:

Packet sniffing is the process of intercepting and analyzing network packets traveling over a network to monitor or troubleshoot traffic.

Algorithm:

1. Get the local host IP address.

2. Create a raw IPv4 socket.

3. Bind the socket to the host IP.

4. Set socket to include IP headers.

5. Enable promiscuous mode to capture all packets.

6. Enter an infinite loop to receive data.

7. Receive raw packet bytes from the socket.

8. Parse the Ethernet frame into dest MAC, src MAC, protocol, and payload.

9. Format MAC addresses and protocol for display.

10. Print the extracted fields and loop to continue sniffing.

Program Code:

```
import socket
import struct
import binascii
import textwrap
def main():
    host = socket.gethostbyname(socket.gethostname())
    print('IP: {}'.format(host))
    conn = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
    conn.bind((host, 0))
    conn.setsockopt(socket.IPPROTO_IP, socket.IP_HDRINCL, 1)
    conn.ioctl(socket.SIO_RCVALL, socket.RCVALL_ON)
    while True:
        raw_data, addr = conn.recvfrom(65536)
        dest_mac, src_mac, eth_proto, data = ethernet_frame(raw_data)
        print('\nEthernet Frame:')
        print("Destination MAC: {}".format(dest_mac))
        print("Source MAC: {}".format(src_mac))
        print("Protocol: {}".format(eth_proto))
def ethernet_frame(data):
    dest_mac, src_mac, proto = struct.unpack('!6s6s2s', data[:14])
    return get_mac_addr(dest_mac), get_mac_addr(src_mac), get_protocol(proto), data[14:]
def get_mac_addr(bytes_addr):
    bytes_str = map('{:02x}'.format, bytes_addr)
    mac_address = ':'.join(bytes_str).upper()
    return mac_address
def get_protocol(bytes_proto):
```

```
    bytes_str = map('{:02x}'.format, bytes_proto)
    protocol = ''.join(bytes_str).upper()
    return protocol
main()
```

Output:

```
C:\Windows\System32>python sniff.py
IP: [redacted]

Ethernet Frame:
Destination MAC: 45:00:00:93:5C:C9
Source MAC: [redacted]
Protocol: C0A8
```

Result:

Thus packet sniffing was implemented using raw sockets in python.